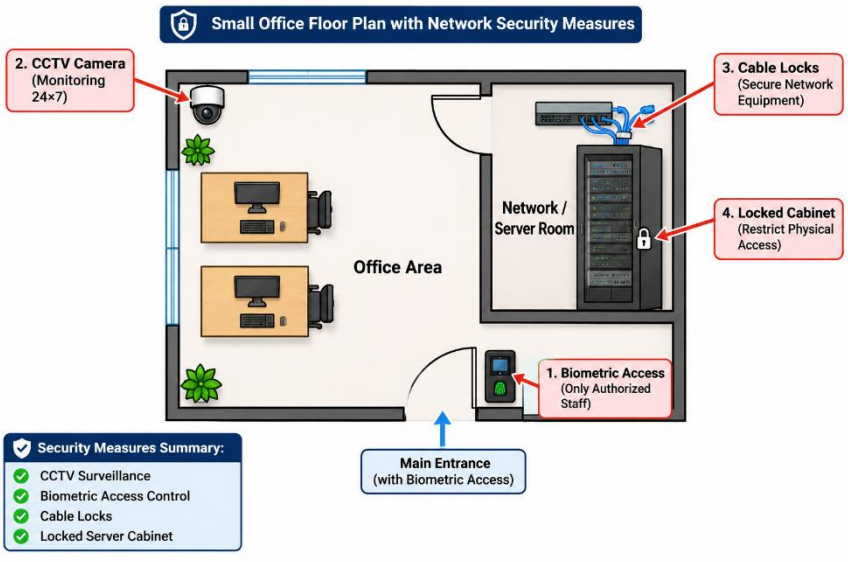
1) Create a table listing 5 common network threats (such as phishing, DDoS, man-in-the-middle, malware, and password attacks), and for each, write one real-world example from an app or service you use (like WhatsApp, Instagram, or Paytm).

No.	Network threat	Description	Real-world example
1	Phishing	Tricking users into revealing passwords, OTPs, or other sensitive information through fake messages or websites.	A user receives a fake Instagram message claiming their account will be suspended and is asked to log in through a fraudulent link.
2	DDoS (Distributed Denial of Service)	Flooding a server with excessive traffic to make a service unavailable to legitimate users.	Attackers flood a payment service's servers with traffic, potentially preventing customers from completing Paytm transactions.
3	Man-in-the-Middle (MITM)	Intercepting or potentially altering communication between two parties.	A user connects to a malicious public Wi-Fi hotspot while accessing WhatsApp Web, potentially exposing unencrypted traffic or redirecting them to a fake login page.
4	Malware	Malicious software designed to steal data, spy on users, or damage devices.	A user installs a fake WhatsApp APK from an unofficial website that contains spyware capable of stealing personal information.
5	Password attacks	Attempts to guess, steal, or reuse passwords to gain unauthorized account access.	An attacker uses passwords leaked from another website to try to access a user's Instagram account. This is known as *credential stuffing*.

2) List 4 physical security risks that could affect a server room hosting a popular app like Zomato or BookMyShow, and suggest one mitigation step for each risk.

No.	Physical security risk	Example	Mitigation step
1	Unauthorized access	An unauthorized person enters the server room and steals or damages servers hosting customer and booking data.	Install biometric access controls and allow entry only to authorized personnel.
2	Fire	An electrical short circuit causes a fire, damaging servers and interrupting app services.	Install fire detection systems and appropriate clean-agent fire suppression systems.
3	Power failure	A power outage shuts down servers, making food ordering or ticket booking unavailable.	Use uninterruptible power supplies (UPS) and backup generators.
4	Flooding or water leakage	A leaking air-conditioning pipe damages server equipment and causes service downtime.	Install water-leak detection sensors and keep servers and critical equipment away from water sources.

3) Draw a simple floor plan (hand-drawn or digital) of a small office and mark at least 3 physical security measures you would implement to protect network equipment (such as CCTV, biometric access, or cable locks).



4) Use ChatGPT or a similar AI tool to generate a checklist of 7 actions a network admin should take to mitigate risks from both network threats and physical security breaches, then review and edit the checklist to make it specific for a college computer lab environment.

College Computer Lab Network Security Checklist

1. Install and update antivirus and endpoint security

- Ensure every lab computer has active antivirus, firewall protection, and automatic security updates. Restrict students from disabling security software.

2. Secure Wi-Fi and network devices

- Use WPA2-AES or WPA3 for Wi-Fi, strong administrator passwords, and separate student, faculty, and administration networks using VLANs.

3. Control user accounts and permissions

- Provide individual student logins, apply least-privilege access, block unauthorized software installation, and disable unused accounts.

4. Back up important data

- Back up student project files, lab configurations, and academic records regularly to a secure location. Test recovery periodically.

5. Install CCTV and monitor the lab

- Place CCTV cameras at lab entrances and near network equipment, following college privacy policies. Restrict and secure access to recordings.

6. Restrict physical access to network equipment

- Keep switches, routers, servers, and patch panels in a locked cabinet or restricted server room. Use authorized access cards or keys and cable locks.

7. Inspect, monitor, and educate

- Review network logs, check for unknown devices, inspect cables and ports, report suspicious activity, and train students not to share passwords or connect unauthorized devices.